

Compact Representations of Human Interception Movements

Low-dimensional models for representing and generating human interception movement

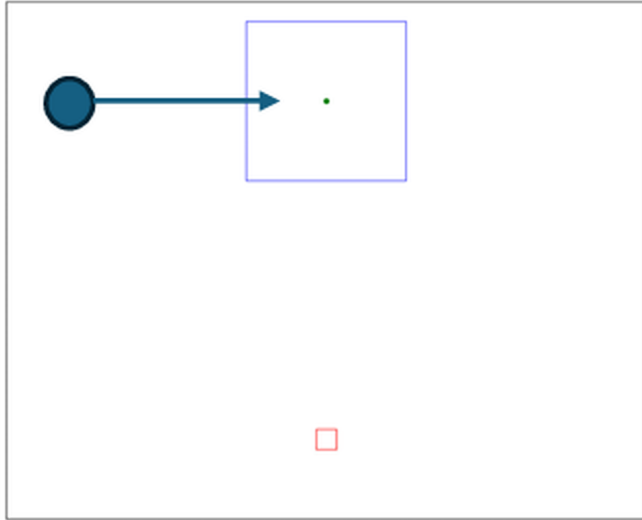
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Workshop on Deep Learning, Tel Aviv University

The experimental task and the data structure

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The experimental task

- **Participant's task:** move the finger from the lower start zone into the upper interception square exactly as the ball passes through it.
- **Conditions:** the free eye-movement condition is analysed. Six target start locations (3 right, 3 left) over a continuous speed range.
- **Sample:** 4,732 valid trials from 28 participants, measured in 3D at 240 Hz with a Polhemus system.

A strict split that prevents information leakage

Four fixed folds: 17 training participants, 4 validation and 7 test. Every participant appears in a test fold exactly once.

What we set out to answer

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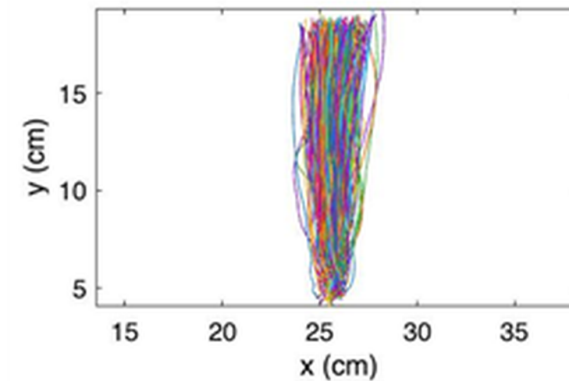
The core question

Human movement is stochastic: perception and motor control vary from trial to trial. One participant, one target - and every trial takes a different path.

Can a compact fingerprint capture that spread?

Compress 200 numbers - the 100×2 path - into a 3-to-8 dimensional fingerprint of an individual's movement distribution.

100×2 coordinates $\longrightarrow$ $n = 3$ or 8 latent numbers (durations predicted)



What would count as success, evaluated on participants the model never trained on

SUCCESS TEST 1

Reconstruct

Rebuild specific recorded trajectories from the compressed code.

SUCCESS TEST 2

Generate

Sample new trajectories matching that person's stochastic distribution.

SUCCESS TEST 3

Stay personal

The fit must come from that person's own fingerprint, not a generic one.

Trajectory processing and the compression architectures

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Stage in the model	Spline + PCA (linear baseline)	Neural models (VAE / CVAE / CAE)
Primary input	The same 100×2 normalised trajectory	The same 100×2 trajectory; CVAE and CAE add 5 task conditions
Where standardisation happens	On the 18 spline coefficients, after fitting	On the 200 coordinates, with train-set constants
Encoder / compression	Cubic B-spline with 5 knots fit, 9 coefficients per axis -> PCA to n latents	Two fully connected 256-unit ReLU layers -> latent n
Decoder / reconstruction	Inverse PCA -> multiply by the spline basis for the full path	Two 256-unit ReLU layers -> separate heads for 200 coordinates and 2 durations
Timing prediction	Separate Ridge regression on the PCA scores + conditions	Dedicated head predicting initiation and movement time

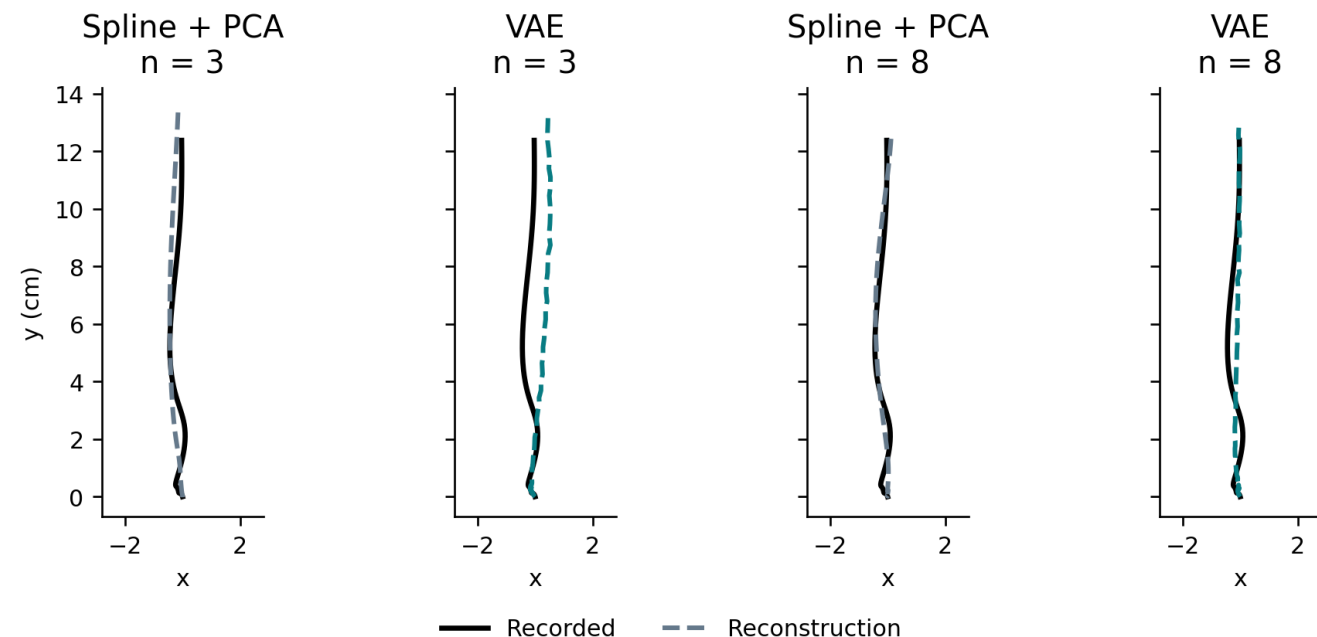
Three neural variants, one architecture

VAE - Variational Autoencoder, trained on the trajectory alone. CVAE - Conditional Variational Autoencoder, which also receives the task conditions. CAE - Conditional Autoencoder, the same network without the variational latent.

All are compared at identical latent dimensions, $n = 3$ (extreme compression) and $n = 8$ (extended capacity).

Task 1: Trajectory Reconstruction

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One median-duration trial. Black: recorded; dashed: reconstructed. Both axes in cm at equal scale.

Trajectory reconstruction error (MSE, cm²)

Model	MSE n=3	MSE n=8
Spline + PCA	0.0972	0.0228
VAE (Unconditional)	0.1483	0.0335
Conditional AE (CAE)	0.1697	0.0260
CVAE	0.1770	0.0325
Condition-only Ridge (Floor)	1.2338	1.2338

Key insight on pointwise reconstruction
PCA wins at reconstructing a single trajectory.

As a linear projection, PCA is analytically optimal for minimising squared error among all linear methods. VAEs trade off pointwise accuracy to obtain a continuous and samplable latent space.

Task 2: generating a personal movement distribution

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How do we generate movements from a fingerprint?

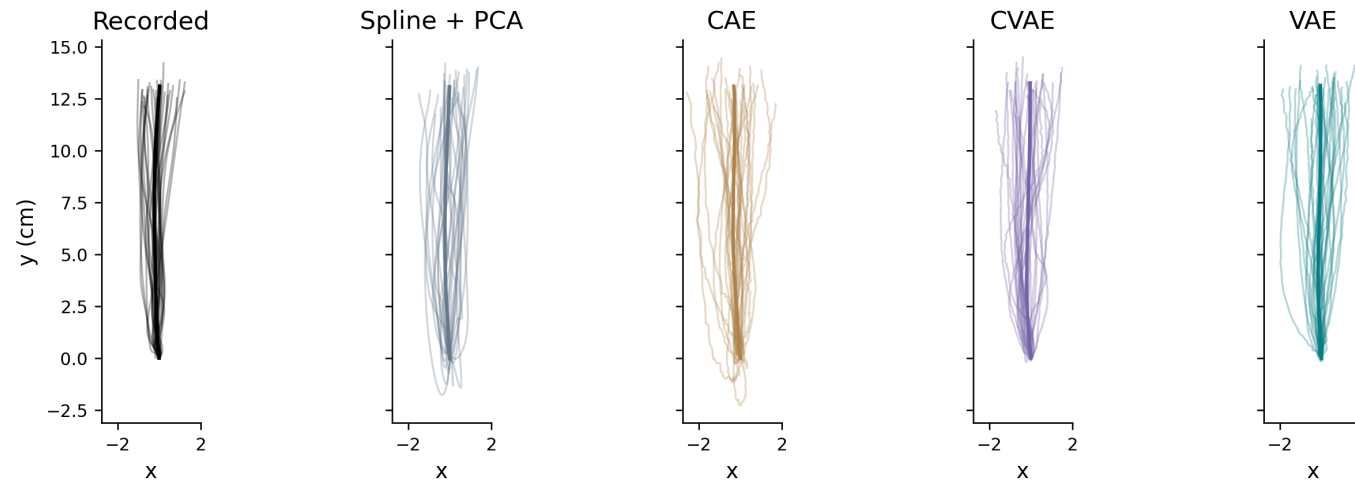
- **Enrollment:** earlier context trials of the participant, whose mean defines the personal fingerprint centre.
- **Stochastic sampling:** draw 120 latent vectors around that centre using a shared covariance learned from training data.
- **Unseen movements:** query trials are held aside to evaluate the distribution and are never exposed to the model.

Below, $n = 8$. Thin: first 30 draws. Thick: their mean. Equal x/y scale, cm.

$$\text{fingerprint centre: } \theta_s = (1 / |C_s|) \sum c_i$$

$$\text{one generated movement: } z \sim N(\theta_s, \Sigma_{\text{train}})$$

c_i is the trial's code: the encoder mean μ_i for the neural families, the PCA scores for spline + PCA.



How we score a generated distribution

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WHAT WE MEASURE

11 kinematic features

Initiation and movement time, peak speed, time to peak, path length, straight-line distance, curvature, maximum lateral deviation and the three endpoint coordinates.

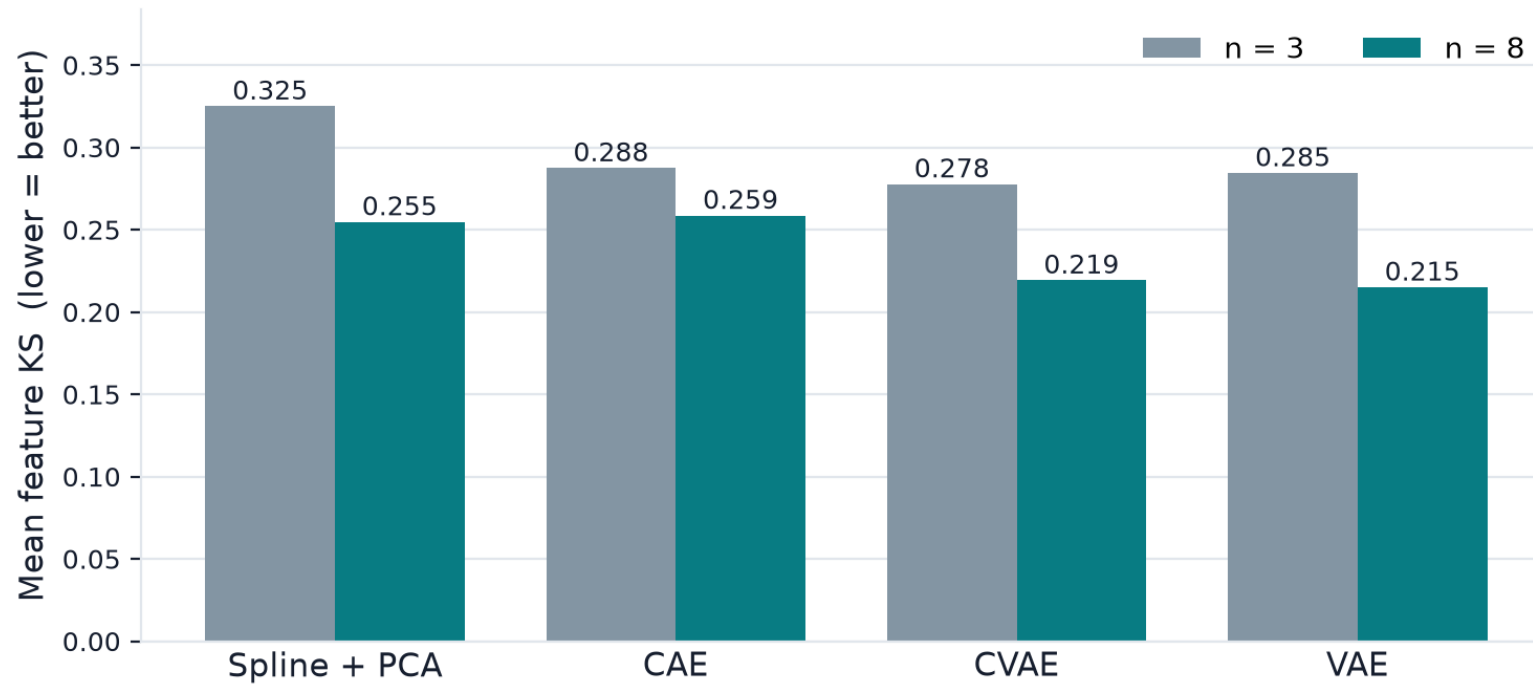
HOW WE COMPARE

Marginal Feature Matching

One KS test per feature, averaged over the 11 features into Mean KS. Lower is better.

Generation results: VAE leads on movement distributions

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A reversal in the ranking: VAE wins at generation

Although Spline + PCA won pointwise reconstruction, VAE and CVAE achieve much better distribution matching (corrected $p < 0.001$ under BH and Holm). VAE at $n = 8$ was set as the dashboard default.

Validating the personal fingerprint (Control Test)

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participants improved markedly

Using the personal fingerprint improved distribution matching against generic attribution ($p < 1.5 \times 10^{-6}$).

Does the personal fingerprint really matter?

To verify that the model is not merely producing a generic average movement, we replaced the fingerprint while the decoder, the noise and the conditions stayed completely frozen:

Injected fingerprint centre	CVAE n=8 (Mean KS)	VAE n=8 (Mean KS)
The participant's own centroid	0.2194	0.2151
Training-population mean	0.2896	0.2841
Another test participant	0.3172	0.3144

Recommended summary: the trade-off between reconstruction

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To reconstruct known trajectories

Spline + PCA

Recommended when the goal is compact compression and the most accurate pointwise reconstruction of a measured trajectory (MSE = 0.0228).

For generation and simulation

VAE (n = 8)

Recommended for simulating a participant's future movements. Produces the kinematic distributions closest to reality (MMD squared = 0.0639).

! Key caveat

Direct measurement from the participant's own context data still beats latent regressions at predicting pointwise summary measures. The compact representation is meant to generate whole trajectories, not a single number.

The applied product: an interactive dashboard

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1

Model and dimension

Supports CAE, CVAE, VAE and Spline + PCA at dimensions 2, 3, 4 and 8.

2

Fingerprint selection

Slider control over the latent coordinates and real-time simulation of the resulting trajectory.

3

Distribution sampling

Generate 25 to 500 trajectories and compare them against the participant's real data.

Streamlit App

Run locally in a terminal:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510
```

Backup: resampling and the two time windows

BACKUP

Model input: target motion to final tracker sample.

Feature input: detected finger onset to end.

A phase of 0.5 means half the selected time interval, not half the path length.

$$\mathbf{P}_j = \mathbf{S}(j/99) - \mathbf{S}(0), \quad j = 0, \dots, 99.$$

100 samples are a common representation, not 100 independent observations.

Backup: what becomes the PCA matrix?

[BACKUP](#)

Object	Dimensions	One row / one column means
Sampled trial	100×2	Time point / x or y
Spline basis B	100×9	Time point / basis function
Trial coefficient vector	18	Nine x coefficients followed by nine y
Training coefficient matrix	$N \text{ training} \times 18$	Trial / coefficient
PCA directions V	$18 \times n$	Coefficient / retained component
Training latent codes	$N \text{ training} \times n$	Trial / latent coordinate

Five interior knots + degree 3 + 1 = nine coefficients per axis.

Backup: fitting and decoding spline + PCA

BACKUP

Fit the two coordinate coefficient vectors by least squares.

Validation selects raw or standardized coefficients.

Inverse PCA gives coefficients; multiplying by B restores the path.

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \oslash \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

m and d combine centering/scaling; V contains retained PCA directions. Raw PCA uses d = 1.

Backup: predicting physical timing

BACKUP

Stage	Spline + PCA	Neural models
Predictors	n PCA scores + 5 conditions	Latent code + conditions where enabled
Targets	Two standardized $\log(t + .001)$ values	Same two standardized targets
Predictor	Ridge, alpha selected on validation	Separate decoder output head
Inverse	Undo scaling, exponentiate, subtract .001	Same inverse transform

Observed timing never enters the trajectory encoder. Reconstruction-time timing prediction sees the observed shape.

Backup: the neural training objective

BACKUP

Coordinate error sums 200 squared differences.

Timing error sums two squared differences and receives weight 20.

KL sums across latent dimensions; the batch averages the total.

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} \mid \mathbf{v}, \mathbf{c}) \parallel \mathcal{N}(\mathbf{0}, I)].$$

Replacing both summed terms by separate MSE means would change the effective timing weight.

Backup: VAE, CVAE and CAE

BACKUP

Property	VAE	CVAE	CAE
Trajectory encoder	Two hidden layers	Two hidden layers	Two hidden layers
Task inputs	Masked to zero	Encoder and decoder	Encoder and decoder
Latent training code	Gaussian sample	Gaussian sample	Deterministic
KL penalty	Yes	Yes	No
Reconstruction code	Posterior mean	Posterior mean	Deterministic code

A nonsignificant conditional-model advantage does not mean the experimental conditions are irrelevant.

Backup: where the generated variation comes from

BACKUP

The personal center averages context codes.

Training residuals estimate within-person latent variation.

Variational models also add average posterior variance.

$$\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}}_{s(i)}) + \text{diag}(\mathbb{E}_{\text{train}}[\boldsymbol{\sigma}_i^2]) + 10^{-6}I.$$

One shared covariance for every participant; 120 draws around each personal center.

Backup: leakage and held-out evaluation

BACKUP

Information	Training	Validation	Test context	Test query
Fit network/PCA/scalers	Yes	No	No	No
Select checkpoint/settings	No	Yes	No	No
Personal center	No	No	Yes	No
Reconstruction input	No	No	No	Observed path
Generation task mixture	No	No	No	Metadata only
Generation scoring	No	No	No	Paths and timings

Population model fitting and participant enrollment are separate stages.

Backup: what the eleven features measure

BACKUP

Group	Features
Timing	Initiation time; movement time; relative time to peak speed
Speed	Peak speed; heuristic speed-peak count
Geometry	Path length; straight-line distance; curvature ratio; maximum lateral deviation
Endpoint	Final x; final y

Generated feature computation uses predicted timing and an onset-based movement crop.

Backup: KS and joint distribution distances

BACKUP

KS: largest gap between two empirical cumulative curves.

Mean KS: average of eleven feature statistics.

Energy and MMD: compare whole vectors and account for joint structure.

$$D_{sf} = \sup_x |F_{sf}(x) - G_{sf}(x)|,$$

$$\overline{D}_s = \frac{1}{11} \sum_{f=1}^{11} D_{sf}.$$

KS statistics are not p-values. Correlated features can affect joint geometry.

Backup: energy discrepancy and mean collapse

[BACKUP](#)

Energy compares cross-set distances with within-set distances.

A generator needs plausible variation as well as a good center.

Example reference	Mean-path RMSE	Path energy
VAE n=8	0.1604	0.0548
Repeated context mean	0.1146	0.4621

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

Lower is better. The mean-only output has a closer mean but a much worse distribution.

Backup: Gaussian-kernel MMD

BACKUP

Kernel: $\exp(-\gamma \times \text{squared Euclidean distance})$.

Gamma comes from the median positive training pair distance.

Within-set diagonals are excluded in the unbiased estimator.

$$\widehat{\text{MMD}}_u^2 = \frac{\sum_{i \neq j} k(X_i, X_j)}{m(m-1)} + \frac{\sum_{i \neq j} k(Y_i, Y_j)}{r(r-1)} - \frac{2 \sum_{i,j} k(X_i, Y_j)}{mr}.$$

Finite-sample unbiased MMD squared may be slightly negative. It is not an accuracy percentage.

Backup: 28 paired scores, not thousands of independent trials

BACKUP

First average neural seeds within each participant.
Then compare paired participant scores using two-sided Pratt Wilcoxon.
Round differences to 12 decimals before ranking.

Correction family	Tests
Original seven endpoints x ten pairs x two dimensions	140
New two path endpoints x ten pairs x two dimensions x two geometries	80
VAE replacement controls	6
CVAE replacement controls	12

BH controls false discoveries under its assumptions. Holm is a familywise-error sensitivity. Nonsignificance is not equivalence.

Backup: the boundaries of the conclusion

BACKUP

Useful same-session personal centers do not establish stable traits.

Direct context summaries beat latent linear probes on all 14 tested targets.

Condition routes work, but their value depends on the task and model.

Minimum-jerk decomposition remains a separate investigation.

The 31-page laboratory report retains the extended results and Jason's open questions.

Backup: launch and source navigation

BACKUP

Repository terminal:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510
```

Browser: <http://127.0.0.1:8510>

Stop: Ctrl+C in that terminal

The standalone dashboard archive contains launch instructions for use without the full research checkout.